Locarda Technical Assessment

Due Date: **3/29/2026 @ 11:59pm EST**

Background

Locarda is a forensic AI platform built for medicolegal death investigation. Our core workflow is voice-to-report — investigators and forensic pathologists dictate findings in the field, and Locarda transcribes, structures, and formats them into complete, template-matched reports. Deviations from the user's report format and/or hallucinations can result in legal scrutiny, so this must be avoided at all costs. At the same time, the platform must generalize to any user.

This project focuses on one of our key report sections: the Evidence of Injury (EOI). Your task is to build a standalone interactive demo of this workflow — from voice input to structured output — that reflects the quality and design standard of our product.

What You're Building

A Next.js web demo that allows a user to:

- Paste in an EOI report template
- Dictate or record findings via voice
- Receive a formatted, structured output that maps to the template

The demo should feel like a real product — not a prototype. It should reflect Locarda's UX: clean, dark, clinical, and minimal. Reference locarda.com for design direction.

Technical Requirements & Specifications

- Framework: Next.js
- Transcription: Deepgram API
- AI formatting: Claude API
- UI: Must match Locarda's design language — reference locarda.com
- The demo should be fully interactive and runnable locally

Some common frameworks to develop AI agents are: LangChain, Claude Agents SDK, Strands, Google Agent ADK, Microsoft Agent SDK. Some of these frameworks support Realtime APIs for voice agents, in case that could be of use.

API keys have been provided in the appendix below. For Claude API, use this for inference & testing for the EOI section formatting task only (not development in your IDE or via Claude Code). We recommend using Claude Sonnet 4.6 for the EOI section formatting; however, you will also be given access to Claude Opus 4.6 and Claude 4.5 Haiku for this task.

Deliverables

1. Working EOI Demo (see appendix for more details)

- User can paste in an EOI template
- User can record or dictate findings
- Output is structured, formatted, and matches the template
- UI matches Locarda's design standard

2. Prompt & Context Write-Up

A short doc explaining your approach to the AI layer:

- How did you structure the prompt and why?
- What context did you pass to the model?
- What did you try that didn't work, and what did you change?

3. Iteration Log

- What you built, what broke, what you changed
- Approaches you tried and moved away from
- What you'd do next with more time

4. AI Coding Session Export

- Full transcript of a coding session with agentic development environment(s)
  - This should be from a feature you built for Deliverable #1

5. Architecture Proposal — Multi-Section Support

Scenario: A user dictates findings across multiple sections of a report in a single session, instead of dictating one section at a time. How would the system know which content belongs to which section?

Write up how you would design this — the agent structure, context flow, how ambiguity between sections would be handled, and any architecture or model choices you'd make. Optional: actually, implementing a prototype and including it as part of your submission.

How to Work

- Treat this like your first two weeks on the team — own it
- Ask questions early, not right before the deadline
- We care about how you think as much as what you ship
- Document your decisions — clarity matters here

- If something is ambiguous, make a call and note why

Submission

- GitHub repo with working Next.js project
- README with setup instructions and how to run locally
- Prompt write-up and architecture proposal as a separate doc or included in the README
- Send to founders@locarda.com by the agreed deadline

Appendix

API Keys:

DEEPGRAM_API_KEY=399c8152c399d5ada342f50d7b6c826311740dd8

ANTHROPIC_API_KEY=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

Evidence of Injury (EOI) Section Formatting Problem Statement

Locarda is a forensic AI platform for medicolegal death investigation. The core workflow is voice-to-report: forensic pathologists dictate findings in the field, and Locarda transcribes, structures, and formats them into complete, template-matched autopsy reports. Deviations from the user's report format or hallucinated findings can result in legal scrutiny and must be avoided. At the same time, the platform must generalize to any user and any office — including pathologists who use entirely different formatting conventions.

This description focuses on the Evidence of Injury (EOI) section: one of the most structurally complex parts of the autopsy report. It documents all traumatic findings in a structured, template-matched narrative derived from raw voice dictation.

What You're Building

A Next.js web demo where a user can:

- Paste in an EOI report template — any format, any office
- Dictate or record findings via voice (Deepgram API for transcription)
- Receive a structured output that matches the pasted template exactly (Claude Sonnet 4.6)
- The demo should feel like a real clinical product — dark, minimal, no prototype energy

Stack: Next.js · Transcription: Deepgram API · Formatting: Claude Sonnet 4.6 · Design reference: locarda.com

The Core Concept: Template-Driven Formatting

Locarda does not enforce a single report format. Every medical examiner office uses a different template. The system's job is to infer the format from one example pasted by the user, then apply it consistently to new dictation — whether it is Kansas-style prose paragraphs, California-style Roman numeral bullet lists, or any other convention.

The AI does not impose structure. It mirrors the structure already defined in the pathologist's pasted template. This is what separates Locarda from a generic transcription tool.

① TEMPLATE INPUT	② VOICE DICTATION	③ STRUCTURED OUTPUT
User pastes a prior EOI section. Any format, any office.	Pathologist dictates. Deepgram transcribes. Raw, messy, with STT errors.	Claude maps transcription to the template's exact structure, layout, and terminology.

Assessment Criteria

- **Template fidelity:** Output mirrors the pasted template's exact structure — headings, bullet vs. prose, body-region taxonomy, and boilerplate phrasing.
- **Injury type classification:** GSW, MBFT, and no-trauma cases are correctly identified and routed to the appropriate subsection structure.
- **Body-region sorting:** Injuries routed to correct regions: Head and neck / Torso / Internal injuries / Extremities (prose) or I–III (CA style).
- **STT error correction:** Common transcription errors silently corrected without changing clinical meaning ('lip' → 'ribs', 'subcalal' → 'subgaleal', 'confused' → 'contused').

- **Mid-dictation corrections:** 'Sorry' / 'actually' corrections applied — the erroneous fragment is discarded.
- **Measurement formatting:** Spoken fractions and compound measurements correctly converted to numeric format.
- **Hallucination avoidance:** Zero fabricated findings — only what was dictated appears in the output.
- **UI/UX quality:** Demo looks and feels clinical — dark, minimal, consistent with Locarda's design language.

How to Read These Examples

Each case contains three panels:

TEMPLATE PLACEHOLDER (purple) — The EOI template style the user would paste into the demo UI. This is the input that drives format — the candidate's system must infer structure from whatever is pasted here. The placeholder text shows the expected structure for that case.

DICTATION INPUT (amber) — Raw voice transcription from Deepgram — including STT artifacts, hesitations, mid-dictation corrections, and spoken numbers/fractions.

EXPECTED OUTPUT (green) — The correctly structured, template-matched EOI section Claude should produce.

Case Index

#	Injury Type	Complexity	Description
1	Gunshot Wound	High	GSW — Head (Near Contact Range, Penetrating)
2	Gunshot Wound	High	GSW — Torso (Two Wounds + Mid-Dictation Correction)
3	Multiple Blunt Force Trauma	High	MBFT — Full Body (20+ Injuries, Multi-Region Sorting)
4	Multiple Blunt Force Trauma	Low	MBFT — Extremities Only, Minimal (STT Error Correction)
5	Multiple Blunt Force Trauma	Low	MBFT — Extremities Only, Measurement Formatting
6	Multiple Blunt Force Trauma	Minimal	MBFT — Single Injury (Boilerplate + Region Classification)
7	None	Minimal	No Trauma — Child Template
8	None	Minimal	No Trauma — Adult Template
9	Multiple Blunt Force Trauma	Medium	MBFT — STT Error Correction (Multiple Errors)

10	Multiple Blunt Force Trauma	Medium	MBFT — Explicit Formatting Instruction + Deduplication
11	Multiple Blunt Force Trauma	Medium	MBFT — Instruction-Style Dictation with Section Tags
12	Multiple Blunt Force Trauma	High	MBFT — CA-Style (Roman Numerals, Bullets, External/Internal)
13	Multiple Blunt Force Trauma	Low	MBFT — CA-Style Head Only + Conversational Noise Filtering

Case 1 GSW — Head (Near Contact Range, Penetrating)

Injury Type: Gunshot Wound	**Complexity:** High

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described with body in anatomical position. Gunshot wounds: [Wound heading]: Wound of entrance: [Location] has a [size] GSW of entrance located [coords]. Wound path: [Anatomical sequence]. Projectile recovery: [Recovered / None]. Wound direction: [Direction].	Penetrating near contact range, puncture length of head, right temple scalp as a quarter inch contour point located three inches in feet to top of the head and three inches right of midline. The one that's associated with circumferential three slash four inch abrasion, some soot, and some black powder, and stippling. Blood exudes from the nose and the right ear.	Described with body in anatomical position. Gunshot wounds: Penetrating near contact range gunshot wound of head: Wound of entrance: Right temple scalp has a 1/4 inch gunshot wound of entrance located 3 inches inferior to top of the head and 3 inches right of midline, associated with a circumferential 3/4 inch abrasion, soot, and stippling. Wound path: Right temporal scalp, inwardly beveled fracture of the right temporal bone with marginal soot, right temporal dura, right temporal lobe, longitudinal fissure, left parietal lobe, dura, externally beveled fracture of the left

Clothing: [Involvement]. Associated injuries: [List]. Multiple blunt force trauma: [Body region]: [Injury description]. Most of the injuries mentioned once will not be repeated.		parietal bone. Projectile recovery: A deformed projectile is recovered from the head, located in the left parietal subgaleal space. Wound direction: Front to back, right to left, and upward. Clothing: Accompanying clothes are not involved. Associated injuries: Diffuse subgaleal hemorrhage; fracture of the calvarium involving the basilar skull, temporal, sphenoid, and occipital bones; lacerated brain along the gunshot wound path; inwardly beveled fracture of the right temporal bone; externally beveled fracture of the left parietal bone. Multiple blunt force trauma: Head and neck: Right lateral eyebrow has a horizontal abrasion. Blood exudes from the nose and the right ear. Extremities: Right knee has an abrasion. Most of the injuries mentioned once will not be repeated.

Examiner notes: *Full GSW taxonomy with all sub-headings (entrance, path, recovery, direction, clothing, associated). Multi-turn session built the complete wound path from incremental dictation.*

Case 2 GSW — Torso (Two Wounds + Mid-Dictation Correction)

Injury Type: Gunshot Wound	**Complexity:** High

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described in anatomical position. [Injury type]: Head and neck: [Findings]. Torso: [Findings]. Internal injuries: [Findings]. Extremities: [Findings]. Most of the injuries mentioned once will not be repeated.	Gunshot wounds of, torso. Contact range gunshot wound. Sorry. Contact range, perforating gunshot wound of right Upper abdomen near the midline has a one eighth inch gunshot wound surrounded with one quarter inch circumferential soot and abrasion. Located 24 inches inferior to the top of the head. Soot and circumferential abrasion are visible. No stippling is visible. Posterior aspect actually, Right side of this wound has a one inch red contusion. Wound path, skin soft tissue, anterior abdominal wall, Mesentery. Small bowel. Thoracic aorta located five centimeter inferior to the branch of left subclavian artery. Thoracic vertebrae on the left costovertebral junction at t seven and exit to the mid back. Wound of exit, Left side of mid back near the midline has a quarter inch exit wound located 16 inches in field to top of the head and quarter inch left of midline. Own direction, front to back, right to left, and upward. Associated injuries, laceration and hemorrhage. Anterior abdominal wall, mesentery, small bowel, thoracic aorta, and	Described in anatomical position. Contact range gunshot wound of right upper abdomen: Wound of entrance: Right upper abdomen near midline has a 1/8 inch gunshot wound surrounded with 1/4 inch circumferential soot and abrasion located 24 inches inferior to the top of the head. Soot and circumferential abrasion are visible. No stippling is visible. Posterior aspect right side of this wound has a 1 inch red contusion. Wound path: Skin, soft tissue, anterior abdominal wall, mesentery, small bowel, thoracic aorta located 5 centimeters inferior to the branch of left subclavian artery, thoracic vertebrae on the left costovertebral junction at T7 Wound of exit: Left side of mid back near midline has a 1/4 inch exit wound located 16 inches inferior to top of head and 1/4 inch left of midline. Projectile recovery:

	peripancreatic soft tissues. Fractured thoracic vertebrae at t seven. Clothing. T shirt has defect anterior corresponding to entrance wound and posterior to exit wound. Contact range. Perforating gunshot wound of left upper abdomen. Wound of entrance. Left upper abdomen has a one eighth inch entrance wound surrounded with circumferential one quarter inch soot and abrasion. Four hyphen one slash two inch left of midline and 24 inches inferior to top of the head. Wound path skin soft tissue. Left anterior abdominal wall. Mesentery. Omentum. Bowel. Left lateral fractured sixth rib and exit to left lateral upper back. Wound of exit. Left lateral upper back one slash two inch exit wound 15 inches inferior to top of head, seven inches left of midline. One direction, front to back, right to left, and upward. Associated injuries, left hemothorax approximately one liter. Actually put this in the previous associated injury. Mesentery omentum bowel lacerated. Left lateral sixth rib is fractured. Projectile recovery, none.	None. Wound direction: Front to back, right to left, and upward. Clothing: T-shirt has defects on the anterior aspect corresponding to the entrance wound and on the posterior aspect corresponding to the exit wound in the midline. Associated injuries: Laceration and hemorrhage of anterior abdominal wall, mesentery, small bowel, thoracic aorta, and peripancreatic soft tissues. Fractured thoracic vertebrae at T7 at the costovertebral junction. Left hemothorax with approximately 1 liter of blood and blood clots. Contact range gunshot wound of left upper abdomen: Wound of entrance: Left upper abdomen has a 1/8 inch entrance wound surrounded with circumferential 1/4 inch soot and abrasion located 4 1/2 inches left of midline and 24 inches inferior to top of head. Wound path: Skin, soft tissue, left anterior abdominal wall, mesentery, omentum, bowel, left lateral fractured sixth rib Wound of exit:

		Left lateral upper back has a 1/2 inch exit wound located 15 inches inferior to top of head and 7 inches left of midline. Projectile recovery: None. Wound direction: Front to back, right to left, and upward. Clothing: T-shirt has wound defects on the anterior and posterior aspects. Associated injuries: Mesentery, omentum, and bowel are lacerated along the wound path. Left lateral sixth rib is fractured. Most of the injuries mentioned once will not be repeated.

Examiner notes: *Two contact-range perforating GSWs. Mid-dictation 'Actually put this in the previous associated injury' correction moves left hemothorax to the first wound's associated injuries.*

Case 3 MBFT — Full Body (20+ Injuries, Multi-Region Sorting)

Injury Type: Multiple Blunt Force Trauma	**Complexity:** High

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this	Multiple blunt force trauma. The face has multiple red brown abrasions on the mid frontal scalp measuring one by one one by three slash four inch and mid forehead measuring one and a half inch in maximum	Described in anatomical position. Multiple blunt force trauma: Head and neck:

case uses --- Described in anatomical position. [Injury type]: Head and neck: [Findings]. Torso: [Findings]. Internal injuries: [Findings]. Extremities: [Findings]. Most of the injuries mentioned once will not be repeated.	dimension. The right medial orbit has abrasions and contusions. Periorbital skin has contusions. Left supraorbital and lateral orbital, skin has abrasions and contusions. The right maxilla has three by one and a half inch red abrasions and contusions. Right lateral maxilla has one by one inch yellow red abrasions. The chin has a red contusion and an small abrasion. Left lateral maxilla and upper maxilla has a one by one inch yellow red abrasion. The left supraorbital skin has a one by half inch yellow red abrasion. The left clavicular area and left lateral neck has a vertical five by one slash two inch area of multiple vertical abrasions and contusions. The right lateral neck has a one by one inch red abrasion. The mid chest has a red abrasion measuring three by two and a half inches. The left upper abdomen has a two by one slash four inch abrasion. The left lateral abdomen also has another one slash four inch abrasion. Left upper extremity has fractured dorsum of left little finger and middle finger sorry. Left little finger and ring finger at the metacarpal joint with overlying exposed lacerated four by two inch	Mid frontal scalp has red brown abrasions measuring 1 x 1 x 3/4 inch and mid forehead measuring 1 1/2 inches in maximum dimension. Right medial orbit has abrasions and contusions. Periorbital skin has contusions. Left supraorbital and lateral orbital skin has abrasions and contusions. Right maxilla has 3 x 1 1/2 inch red abrasions and contusions. Right lateral maxilla has 1 x 1 inch yellow red abrasions. Chin has a red contusion and a small abrasion. Left lateral maxilla and upper maxilla has a 1 x 1 inch yellow red abrasion. Left supraorbital skin has a 1 x 1/2 inch yellow red abrasion. Left clavicular area and left lateral neck has a vertical 5 x 1/2 inch area of multiple vertical abrasions and contusions. Right lateral neck has a 1 x 1 inch red abrasion. Torso: Mid chest has a red abrasion measuring 3 x 2 1/2 inches. Left upper abdomen has a 2 x 1/4 inch abrasion. Left lateral abdomen has a 1/4 inch abrasion. Internal injuries: Brain has subarachnoid hemorrhage. Right anterior ribs 1 through 5 are fractured. Sternum is fractured. Left posterior ribs 6 and 7 are fractured.

	skin. The palmar aspect of the left little finger also has a large laceration with completely nearly totally fractured metacarpal joint. The extensor aspect of the left fore wrist sorry, forearm over the wrist has a three by three slash four inch laceration. The left extensor aspect of left upper forearm and left elbow have multiple red abrasions. Left lateral pelvis has a three slash four inch yellow red abrasion. The left lower extremity is externally rotated due to fracture of the left sorry, is externally rotated. The left knee has multiple abrasions and contusions. Anterior aspect of the left leg has a three by one inch abrasion. The left lateral leg has, confused abrasions. Right anterior ribs one through five are fractured. Sternum is fractured. Left posterior ribs six and seven are fractured. Brain has subarachnoid hemorrhage.	Extremities: Left upper extremity has fractured dorsum of left little finger and ring finger at the metacarpal joint with overlying exposed lacerated 4 x 2 inch skin. Palmar aspect of the left little finger has a large laceration with completely nearly totally fractured metacarpal joint. Extensor aspect of left forearm over the wrist has a 3 x 3/4 inch laceration. Left extensor aspect of left upper forearm and left elbow have multiple red abrasions. Left lateral pelvis has a 3/4 inch yellow red abrasion. Left lower extremity is externally rotated. Left knee has multiple abrasions and contusions. Anterior aspect of the left leg has a 3 x 1 inch abrasion. Left lateral leg has contused abrasions. Most of the injuries mentioned once will not be repeated.

Examiner notes: *20+ injuries sorted across Head and neck / Torso / Internal injuries / Extremities. Mid-dictation 'sorry' corrections are silently discarded. 'confused abrasions' → 'contused abrasions'.*

Case 4 MBFT — Extremities Only, Minimal (STT Error Correction)

Injury Type: Multiple Blunt Force Trauma	**Complexity:** Low

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described with body in anatomical position. Gunshot wounds: [Wound heading]: Wound of entrance: [Location] has a [size] GSW of entrance located [coords]. Wound path: [Anatomical sequence]. Projectile recovery: [Recovered / None]. Wound direction: [Direction]. Clothing: [Involvement]. Associated injuries: [List]. Multiple blunt force trauma: [Body region]: [Injury description]. Most of the injuries mentioned once will not be repeated.	Right wrist has red purple contusions. Right inguinal area has needle punctures. The left inguinal area has major punctures and abrasions.	Described with body in anatomical position. Multiple blunt force trauma: Extremities: Right wrist has red-purple contusions. Right inguinal area has needle punctures. Left inguinal area has multiple punctures and abrasions.

Examiner notes: *'red purple' → 'red-purple' (hyphenation). 'major' → 'multiple' (STT error correction).*

Case 5 MBFT — Extremities Only, Measurement Formatting

Injury Type: Multiple Blunt Force Trauma	**Complexity:** Low

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described in anatomical position. [Injury type]: Head and neck: [Findings]. Torso: [Findings]. Internal injuries: [Findings]. Extremities: [Findings]. Most of the injuries mentioned once will not be repeated.	Left lateral elbow has a one hyphen one slash four inch red contusion. The right elbow has a red purple contusion covering an area two and a half by two inches. Extensor aspect of the right forearm also has multiple red contusions, which are indistinct. The	Described in anatomical position. Blunt force trauma: Extremities: Left lateral elbow has a 1-1/4 inch red contusion. Right elbow has a red-purple contusion covering an area 2-1/2 x 2 inches. Extensor aspect of the right forearm has multiple indistinct red contusions. Most of the injuries mentioned once will not be repeated.

Examiner notes: *Spelled-out fractions → numeric. Trailing incomplete word ('The') dropped. 'red purple' → 'red-purple'.*

Case 6 MBFT — Single Injury (Boilerplate + Region Classification)

Injury Type: Multiple Blunt Force Trauma	**Complexity:** Minimal

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described in anatomical position. [Injury type]: Head and neck: [Findings]. Torso: [Findings]. Internal injuries: [Findings]. Extremities: [Findings]. Most of the injuries mentioned once will not be repeated.	Blunt force trauma left lateral pelvis has an abrasion.	Described with body in anatomical position. Blunt force trauma: Torso: Left lateral pelvis has an abrasion. Most of the injuries mentioned once will not be repeated.

Examiner notes: *Bare-minimum single finding. 'Pelvis' → Torso (not Extremities). Boilerplate wrapper always applied.*

Case 7 No Trauma — Child Template

Injury Type: None	**Complexity:** Minimal

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format	No trauma.	Distinct injuries are not readily visible.

from whatever you paste. --- Example structure this case uses --- Child template — no trauma: Distinct injuries are not readily visible.		

Examiner notes: *Child template phrasing. Template-specific boilerplate must match exactly.*

Case 8 No Trauma — Adult Template

Injury Type: None	**Complexity:** Minimal

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Adult template — no trauma: No significant traumatic injuries are identified.	For the Evidence of Injury section: No trauma.	No significant traumatic injuries are identified.

Examiner notes: *Same dictation intent as Case 7 — different output because a different template was pasted.*

Case 9 MBFT — STT Error Correction (Multiple Errors)

Injury Type: Multiple Blunt Force Trauma	**Complexity:** Medium

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described in anatomical position. [Injury type]: Head and neck: [Findings]. Torso: [Findings]. Internal injuries: [Findings]. Extremities: [Findings]. Most of the injuries mentioned once will not be repeated.	Blunt force trauma. The, right leg has few abrasions and green purple contusions. The left leg has, contusion and red abrasions. The dorsum of the right hand has a abraded, one slash four inch abrasion. The abdomen has green contusions on the right side. The chest has, green purple contusion on the right nipple. The forehead has a one hyphen one slash two inch laceration. The left lateral elbow, left lateral forearm, dorsum of the left hand have multiple lacerations, contusions, and abrasions covering an area 14 by four inches. Multiple stones are on the back. Left knee has a vertical, hypopigmented linear surgical scar. Sternum is fractured with hemorrhagic intercostal muscles involve involving bilateral anterior ribs two through four. Right lateral lip five and six are fractured and hemorrhagic. No lobe of right lung is confused. Mid frontal scalp has subcalal hemorrhage	Described in anatomical position. Blunt force trauma: Head and neck: Forehead has a 1-1/2 inch laceration. Mid frontal scalp has subgaleal hemorrhage. Torso: Chest has green-purple contusion on the right nipple. Abdomen has green contusions on the right side. Internal injuries: Sternum is fractured with hemorrhagic intercostal muscles involving bilateral anterior ribs 2 through 4. Right lateral ribs 5 and 6 are fractured and hemorrhagic. Right lung is contused. Extremities: Right leg has few abrasions and green-purple contusions. Left leg has contusion and red abrasions. Dorsum of the right hand has an abraded 1/4 inch abrasion. Left lateral elbow, left lateral forearm, and dorsum of the left hand have multiple lacerations, contusions, and abrasions covering an area 14 x 4 inches. Left knee has a vertical hypopigmented linear surgical scar. Multiple stones are present on the back.

		Most of the injuries mentioned once will not be repeated.

Examiner notes: *STT errors: 'lip five and six' → 'ribs 5 and 6'; 'No lobe of right lung is confused' → 'Right lung is contused'; 'subcalal' → 'subgaleal'.*

Case 10 MBFT — Explicit Formatting Instruction + Deduplication

Injury Type: Multiple Blunt Force Trauma	**Complexity:** Medium

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described in anatomical position. [Injury type]: Head and neck: [Findings]. Torso: [Findings]. Internal injuries: [Findings]. Extremities: [Findings]. Most of the injuries mentioned once will not be repeated.	Left knee has a one slash two inch scabbed abrasion. The, right occipital scalp has a one slash two inch laceration surrounded by two inch purple purple contusion. The right forearm has skin slippage. Right parietal occipital scalp has subgaleal hemorrhage and the calvarium has subperiosteal hemorrhage in the underlying bone. Please put measurements in fractions.	Described in anatomical position. Blunt force trauma: Head and neck: Right occipital scalp has a 1/2 inch laceration surrounded by a 2 inch purple contusion. Right parietal occipital scalp has subgaleal hemorrhage and the calvarium has subperiosteal hemorrhage in the underlying bone. Extremities: Left knee has a 1/2 inch scabbed abrasion. Right forearm has skin slippage. Most of the injuries mentioned once will not be repeated.

Examiner notes: *Explicit formatting preference ('Please put measurements in fractions') applied. 'purple purple' → 'purple' (deduplication).*

Case 11 MBFT — Instruction-Style Dictation with Section Tags

Injury Type: Multiple Blunt Force Trauma	**Complexity:** Medium

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- Described in anatomical position. [Injury type]: Head and neck: [Findings]. Torso: [Findings]. Internal injuries: [Findings]. Extremities: [Findings]. Most of the injuries mentioned once will not be repeated.	for the Evidence of Injury section, include the following: Multiple blunt force trauma. Left frontal scalp has a two inch laceration. Right side of neck has a four inch abrasion. Left foot has a laceration. Right femur is fractured with open laceration. Lower back has a large abrasion. The heart is, lacerated. Use section notes: Blunt Force, MBFT - Head injuries, MBFT - Neck injuries, MBFT - Torso injuries, MBFT - Extremity injuries.	Described in anatomical position. Multiple blunt force trauma: Head and neck: Left frontal scalp has a 2 inch laceration. Right side of neck has a 4 inch abrasion. Torso: Lower back has a large abrasion. Internal injuries: Heart is lacerated. Extremities: Left foot has a laceration. Right femur is fractured with open laceration. Most of the injuries mentioned once will not be repeated.

Examiner notes: *Section tags are routing hints only — not reproduced in the output. Instruction preamble is discarded.*

Case 12 MBFT — CA-Style (Roman Numerals, Bullets, External/Internal)

Injury Type: Multiple Blunt Force Trauma	**Complexity:** High

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format from whatever you paste. --- Example structure this case uses --- **MULTIPLE BLUNT FORCE INJURIES** **I. HEAD AND NECK** *External:* • [Injury], [location], [size] • [Injury], [location], [size] *Internal:* • [Injury], [location], [size] **II. THORAX AND PELVIS** *External:* • [Injury], [location], [size] *Internal:* • [Injury], [location], [size] **III. EXTREMITIES** *External:*	Multiple blunt force injuries make three sections with head and neck, thorax and pelvis, and extremities. Subheadings in numeral format divided into external and internal. Head externally: stapled surgical incision right parietal scalp four inches, healing sutured incision right frontal scalp one inch. Contusion frontal scalp four and a half by two inches. Laceration pinna of left external ear one inch. Linear abrasion left cheek three inches. Laceration left parietal scalp one inch. Laceration left posterior parietal scalp two inches. Contusion left lower lip mucosa one inch. Thorax and pelvis externally: right lateral buttocks eight by four inch cluster of contusion and abrasion. Contusions both buttocks twelve by eight inches. Thorax internal: multiple rib fractures left posterior five through eight. Extremities externally: right leg shortened. Right ankle internally rotated palpable crepitus. Right foot edematous. Contusion right lateral ankle five by two inches. Abrasion right medial ankle two inches. Scattered abrasions right	**MULTIPLE BLUNT FORCE INJURIES** **I. HEAD AND NECK** *External:* • Contusion, frontal scalp, 4.5 x 2 inches • Laceration, pinna of left external ear, 1 inch • Abrasion (linear), left cheek, 3 inches • Laceration, left parietal scalp, 1 inch • Laceration, left posterior parietal scalp, 2 inches • Contusion, left lower lip mucosa, 1 inch • Healing sutured incision, right frontal scalp, 1 inch • Stapled surgical incision, right parietal scalp, 4 inches *Internal:* **II. THORAX AND PELVIS** *External:* • Contusions and abrasions (cluster), right lateral buttock, 8 x 4 inches • Contusions (cluster), bilateral buttocks, 12 x 8 inches *Internal:* • Fractures (multiple), left posterior ribs 5-8

• [Injury], [location], [size] *Internal:*	medial leg. Contusion left medial leg three inches. Contusion left medial thigh two inches. Contusion right anterior thigh one half inch. Contusion left upper arm two inches. Incised wound left posterior hand three inches. Abrasion left posterior hand one half inch.	**III. EXTREMITIES** *External:* • Shortening, right leg • Internal rotation with palpable crepitus, right ankle • Edema, right foot • Contusion, right lateral ankle, 5 x 2 inches • Abrasion, right medial ankle, 2 inches • Abrasions (scattered), right medial leg • Contusion, left medial leg, 3 inches • Contusion, left medial thigh, 2 inches • Contusion, right anterior thigh, 1/2 inch • Contusion, left upper arm, 2 inches • Incised wound, left posterior hand, 3 inches • Abrasion, left posterior hand, 1/2 inch *Internal:*

Examiner notes: *CA-style: Roman numeral body-region headers, External/Internal subsections, comma-delimited bullet (injury type, location, measurement). Completely different structure from Kansas prose — the pasted template drives the entire output format.*

Case 13 MBFT — CA-Style Head Only + Conversational Noise Filtering

Injury Type: Multiple Blunt Force Trauma	**Complexity:** Low

TEMPLATE PLACEHOLDER	DICTATION INPUT	EXPECTED REPORT OUTPUT
[PASTE YOUR EOI TEMPLATE HERE] The system will infer your format	What's the one that collapsed like with walking his dog? Contusion, right posterior scalp, four inches. Abrasion healing left	**Head** Contusion, 4 inches, right posterior scalp.

from whatever you paste. --- Example structure this case uses --- **MULTIPLE BLUNT FORCE INJURIES** **I. HEAD AND NECK** *External:* • [Injury], [location], [size] • [Injury], [location], [size] *Internal:* • [Injury], [location], [size] **II. THORAX AND PELVIS** *External:* • [Injury], [location], [size] *Internal:* • [Injury], [location], [size] **III. EXTREMITIES** *External:* • [Injury], [location], [size] *Internal:*	eyebrow one quarter inch. Contusion, right cheek, two inches. Contusion, right frontal scalp, one half inch.	Contusion, 1/2 inch, right frontal scalp. Contusion, 2 inches, right cheek. Abrasion (healing), 1/4 inch, left eyebrow.

Examiner notes: *Opening conversational question filtered out. Injuries anatomically reordered (scalp → cheek → eyebrow). CA format: injury type, measurement, location.*